

**GITAM (Deemed to be University)**  
**24CSEN2011**  
**GSB/GSHSS/GSS/GSCE/GSCSE Degree Examination**  
**IV Semester(Sessional -II)**  
**OPERATING SYSTEMS**  
(Effective from 2025-26 onwards)

**Max. Marks: 45**

**Time: 2 hours**

**Instruction:**

1. All parts of the sections must be answered in one place only.

| <b>Section-A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | <b>Marks</b> |
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| <b>Answer all Questions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |              |
| 1. What is Semaphore, how is Semaphore used as general synchronization tool?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | 2            |
| 2. Define Swapping in memory management?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  | 2            |
| 3. Differentiate between domain and access rights in the context of protection mechanisms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  | 2            |
| <b>Section-B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | <b>Marks</b> |
| <b>Answer all Questions</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |              |
| 4. a. Imagine a small office with two employees, Donald and Benjamin, who share a single high-end photo printer. Because the printer is expensive, only one person can use it at a time<br>To coordinate, they use a whiteboard with two markers and a "Priority Note":<br>The Flags: There are two checkboxes on the board: "Donald wants to print" and "Benjamin wants to print."<br>The Priority Note: A space that says "Current Turn."<br>The Protocol: When Donald wants to print, he marks his checkbox and—to be polite—sets the "Current Turn" to Benjamin. He then waits until either Benjamin checkbox is empty OR it is his turn. Benjamin does the exact same thing.<br>The Question: How does this "whiteboard protocol" (Peterson's Solution) ensure that they never print at the same time, that the printer doesn't sit idle if someone wants to use it, and that Donald doesn't get stuck waiting forever while Benjamin prints 100 copies?                                        |  | 5            |
| <b>(OR)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |              |
| b. A high-end club has an Exclusive VIP Lounge that only has room for one person at a time. Outside the lounge door, there is a specialized digital turnstile. This turnstile has a single memory slot that holds a value: either 0 (Unlocked) or 1 (Locked). There are N guests (Threads) constantly trying to enter the lounge, spend a few minutes inside (Critical Section), leave, and then immediately try to get back in.<br>The Entry Mechanism (Test-and-Set):<br>Each guest uses a "swipe-and-check" move. In one single, unbreakable action (atomic), the guest: Swipes their card to see what the current value is. Simultaneously, it sets the value to 1. If the value they "saw" was 0, they enter. If the value they "saw" was 1, it means someone else was already inside, so they must keep swiping their card repeatedly until they catch a 0.<br>The Question:<br>a. Sequence: Describe the sequence of acquisition for these N guests. Is it "fair" (first-come, first-served)? |  | 5            |



- b. Re-entry: Can a guest who just left the lounge immediately re-acquire the lock before others who have been waiting longer?
- c. Prevention: If re-entry is possible, how do we modify the system to ensure everyone gets a fair turn (Bounded Waiting)?
5. a. Define the dynamic storage allocation problem and evaluate its primary strategies. 5
- (OR)
- b. Define virtualization and demonstrate the conceptual relationship between virtual and physical memory. 5
6. a. Illustrate the process of transforming I/O requests into hardware-level operations with suitable examples. 5
- (OR)
- b. A system has page size 4 KB, process size 40 KB, and physical memory 16 KB. Calculate: 5
- Number of pages
  - Pages in memory
  - Pages in swap space

### Section-C

Marks

**Answer all Questions**

7. a. Identify and explain the three classic problems of synchronization. Provide a solution for at least one using semaphores. 8
- (OR)
- b. Analyze the Readers-Writers Problem in the context of process synchronization. Discuss its constraints, potential variations regarding priority, and illustrate a solution using synchronization primitives. 8
8. a. A system administrator observes that when a large number of processes are running concurrently on a server, the system's performance significantly degrades. Upon further investigation, the administrator identifies that the translation of virtual addresses to physical addresses is taking a considerable amount of time. 8
- Illustrate the mechanism of translation lookaside buffer (TLB) in virtual memory systems with a neat diagram.
  - Based on the TLB diagram, determine how the system's effective access time (EAT) can be calculated to analyze the impact of address translation on the overall system performance.
- (OR)
- b. Illustrate the page-replacement algorithms 8
- FIFO
  - Optimal Page Replacement
- use the reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 for a memory with three frames.
9. a. A modern operating system deployed on a high-performance server experiences I/O bottlenecks when handling multiple concurrent requests from different applications. Critically evaluate the design challenges in the kernel I/O subsystem under such conditions. Also, suggest suitable improvements to enhance performance and scalability. 8
- (OR)
- b. A disk has request queue 90, 170, 44, 139, 10, 125, 68, 35, initial head 50. Perform SCAN, C-SCAN, and C-LOOK. Compute total head movement and compare performance. 8

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